

Jayacharan R

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Education

Vellore Institute Of Technology, *B.Tech in Computer Science (Data Science) – Vellore* | **CGPA: 9.27** July 2023 – present
Achyuta Public School, *CBSE, High School in Bio-Mat – Dindigul, TN* | **AISSCE: 89.6%, AISSE: 98%** June 2017 – Apr 2023

Experience

AI-ML Intern, *Tata Consultancy Services - TCS – Chennai, TN, India* June 2026 – July 2026

- Engineered high-accuracy RAG search (<3.5s latency, 390 tok/s) with FastAPI, LangChain, ChromaDB & Groq.
- Secured pipeline against prompt injections with <15ms overhead by implementing Aegis LLM guardrails.
- Orchestrated real-time Langfuse telemetry across 100% of traces to optimize performance & token costs.

SAP Analytics Cloud Intern, *Blueprint Technologies Pvt Ltd – Bengaluru, KA, India* May 2025 – July 2025

- Delivered 2 SAP Analytics Cloud dashboards (20+ visuals) via Exploratory Data Analysis (EDA), cleaning, live Google Drive sync
- Strengthened insights via 3 predictive models (regression, classification, time-series) & dimensional modeling
- Resolved lat-long ID conflicts, boosting geo-mapping accuracy across 210 countries in SAP Analytics Cloud

Projects

WikiMind: End-to-End Advanced Agentic RAG & Knowledge Engine June 2026 – present

- Boosted Recall@K by 30% across 6.8M articles (21GB+) via LangGraph Hybrid RAG (Dense + BM25 + Rerank).
- Reduced staleness to <5 min & boosted query accuracy by 25% using Redis Knowledge Graphs & spaCy.
- Elevated answer accuracy to 88% at <800ms latency by tuning Self-RAG state machines & NeMo.

Clean-Backup: Intelligent Media Automation & Cloud Sync Engine Jan 2026 – present

- Accelerated deduplication by 10× (1,000+ images/s) & cut storage by 81% via Rust pHash (PyO3).
- Integrated Google Drive sync across 64 REST API endpoints via OAuth2 and a 3-stage Docker build.
- Automated atomic rollback for 50+ batch operations across 14 sessions using journaled SQLite.

Awards

Winner - Roboverse Ideathon 2025, IEEE Robotics and Automation Society, VIT, Vellore Jan 2025

- Architected AI surveillance prototype using YOLO (Ultralytics) for real-time anomaly & movement detection
- Strengthened crime-prevention capability via automated alerts, behavioral analysis, and event logging

Skills

Programming & Frameworks: Python, Java, C, C++, Bash, YAML, HTML, CSS, JS, Pillow, FFmpeg, Tkinter

AI/ML Frameworks: PyTorch, LangChain, Langfuse, Retrieval-Augmented Generation (RAG), Large Language Models (LLM) Observability, LLM Guardrails, Quantization, Fine-tuning, HuggingFace Embeddings, ONNX Runtime, Qdrant (Vector DB), Redis, NetworkX (Knowledge Graphs), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, NVIDIA NeMo Guardrails

Software: SAP Analytics Cloud, Git Version Control, AWS Architecture, Docker Deployment, Arch Linux

Soft Skills: Problem Solving, Team Collaboration, Adaptability, Time Management, Critical Thinking

Certifications

AWS Certified Solutions Architect – Associate Dec 2025

Microsoft Certified: Azure AI Engineer Associate - AI 102 Nov 2025

Microsoft Certified: Azure Data Fundamentals - DP 900 Oct 2025